

CAMERON BASHAM

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer focused on AI systems, simulation telemetry, and GPU-accelerated ML infrastructure. Built real-time vehicle instrumentation in CARLA and CUDA-accelerated CNN training pipelines with structured performance validation. Expected December 2026.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, Lua

AI / ML: PyTorch, CUDA, CNN training, GPU acceleration, Top 1/Top 5 accuracy evaluation

Backend: Spring Boot, REST APIs, JWT/OAuth2, RBAC, Microservices

Cloud: AWS (RDS, VPC), Docker, Git/GitHub

Databases: MariaDB, MySQL

TECHNICAL PROJECTS

CARLA Observability Toolkit | Python

- Designed telemetry extraction framework enabling real-time vehicle-state instrumentation in CARLA.
- Engineered structured pipelines for simulation metrics and validation workflows.
- Integrated sensor data processing for experiment-driven performance analysis.

GPU-Accelerated CNN Training | Python, PyTorch, CUDA

- Developed CUDA-enabled CNN training pipeline with explicit GPU device management.
- Tuned learning rate, batch size, and model depth to improve convergence and reduce overfitting.
- Tracked Top 1/Top 5 accuracy across epochs to analyze training stability and model confidence.

Car Rental Microservices | Java, Spring Boot, AWS

- Built secure REST APIs with JWT/OAuth2 authentication and role-based access control (RBAC).
- Integrated AWS RDS and Docker for managed persistence and containerized development.
- Applied modular microservice design principles for separation of concerns.

PROFESSIONAL EXPERIENCE

Software Developer | Ferris Reservation Tool | 2025–2025

- Designed secure Spring Boot APIs with JWT-based authentication.
- Engineered MariaDB/MySQL schemas ensuring data integrity and access control.
- Developed React frontend components and managed deployment workflows.

Project Manager / Developer | Battlefield 3 Reality Mod | 2021–2024

- Managed issue tracking and validation workflows for distributed contributors.
- Implemented Lua-based feature enhancements and structured QA testing cycles.
- Executed build verification procedures for development and production releases.

U.S. ARMY — CAVALRY SCOUT (E5) | 2012–2019

Non-Commissioned Officer responsible for small-team leadership, mission planning, and equipment accountability; developed strong ownership, accountability, and reliability mindset applied to engineering systems.

EDUCATION

Ferris State University — B.S. Software Engineering | Expected December 2026